L_RAG: A Controlled Ablation Study of Vietnamese Legal Retrieval-Augmented Generation

Vu Khac Binh (23127029) **Van Thi Diem My (23127091)**
Tran Tuan Kiet (23127215) Nguyen Le Thien Ly (23127223)
Nguyen Phuong Thao (23127306)
Faculty of Information Technology
VNUHCM–University of Science

Abstract

Legal question answering requires evidence that is semantically relevant, structurally complete, temporally appropriate, and traceable to an authoritative source. We present L_RAG, a Vietnamese legal Retrieval-Augmented Generation pipeline that combines dense E5 retrieval, BM25, a knowledge-graph overlay, rank fusion, cross-encoder reranking, and citation-grounded generation. On a 500-question benchmark, we find that metadata-enriched embeddings improve document-level success but not exact chunk ranking; the tested hybrid and graph variants reduce retrieval coverage; and cross-encoder reranking consistently improves ranking at a measurable latency cost. Jina gives the strongest fixed-candidate retrieval result, whereas BGE gives the highest lexical answer-overlap scores and mMiniLM is the fastest reranker. In a completed generator-reasoning comparison, GPT-4o-mini chain-of-thought prompting lowers lexical overlap under fixed retrieved context while improving only template-based answerability diagnostics and adding latency. Human-validated legal correctness and citation faithfulness are not available, so the conclusions are bounded to deterministic retrieval, overlap, abstention, and recorded latency diagnostics.

1 Introduction

Legal Information Retrieval (IR) and Question Answering (QA) represent critical components in modern legal technology, enabling legal professionals, citizens, and researchers to efficiently navigate vast repositories of statutes, decrees, circulars, and judicial precedents. However, applying general-purpose Retrieval-Augmented Generation (RAG) models to legal domains presents distinct domain-specific challenges. Unlike open-domain Wikipedia-style QA, legal reasoning demands strict grounding in authoritative statutory context, precise citation of applicable legal provisions (e.g., Articles, Clauses, Items), and strict temporal validity verification (ensuring that superseded or annulled laws are not cited).

1.1 Problem Statement

Conventional RAG pipelines rely heavily on single-stage dense vector retrieval powered by pretrained embedding models. While dense retrieval captures high-level semantic intent, it suffers from three core limitations when deployed on complex Vietnamese legal corpora:

1. **The Semantic-Statutory Vocabulary Gap:** Natural language user queries (e.g., *"Tôi bị phạt bao nhiêu tiền khi không đội mũ bảo hiểm?"*) differ significantly in terminology from legal codifications (e.g., *"Điều 6 Nghị định 100/2019/NĐ-CP: Phạt tiền từ 300.000 đồng đến 400.000 đồng..."*). Pure vector search often maps natural queries to general descriptions rather than the exact punitive provision.

2. **Structural and Hierarchy Blindness:** Legal documents follow strict hierarchical compositions (Law $\rightarrow$ Chapter $\rightarrow$ Article $\rightarrow$ Clause $\rightarrow$ Item). Standard fixed-length text chunking fragments these structural relationships, causing vector search to retrieve isolated clauses missing their governing article definitions or parent law scope.

3. **Temporal and Inter-Document Citation Complexity:** Legal provisions do not exist in isolation; they amend, replace, repeal, or reference other statutory provisions across multiple legal instruments. Pure semantic similarity fails to account for statutory precedence (e.g., Constitution $\succ$ Law $\succ$ Decree) or dynamic validity timelines.

To resolve these challenges, a legal RAG system must unify dense semantic matching, exact sparse keyword matching, statutory structure traversal, and fine-grained cross-encoder reranking.

1.2 Objectives

The primary goal of the **L_RAG** project is to design, implement, and analyze a novel, robust Knowledge-Graph-Augmented Hybrid RAG framework specifically engineered for Vietnamese legal corpora. The specific technical objectives are:

- **Architect an End-to-End Hybrid RAG Pipeline:** Combine dense vector retrieval (`intfloat/multilingual-e5-large`), sparse tokenized keyword search (BM25), and Reciprocal Rank Fusion (RRF) to maximize recall across diverse query types.
- **Develop a Knowledge Graph (G-LRAG) Overlay Module:** Construct an in-memory legal knowledge graph that models containment hierarchies (`DOCUMENT_HAS_PROVISION`, `PROVISION_HAS_CHUNK`) and structural reading orders (`CHUNK_NEXT`, `PROVISION_NEXT`) to dynamically supply document filtering whitelists and same-provision context expansions.
- **Integrate Advanced Multi-Reranker Pipelines:** Deploy and evaluate local Cross-Encoder models—including `mMiniLMv2`, `BGE-Reranker-v2-m3`, `Jina-Reranker-v2`, and `Qwen3-Reranker-0.6B`—to refine candidate rankings prior to LLM generation.
- **Construct a Standardized 500-Query Vietnamese Legal Benchmark:** Curate a comprehensive benchmark of 500 legal QA pairs covering answerable, unanswerable, single-hop, multi-hop, citation, cross-document, and legal validity categories.

1.3 Contributions

The key contributions of this work include:

1. **Comprehensive System Implementation:** We deliver a production-ready, modular codebase supporting end-to-end legal data ingestion, vector indexing, knowledge graph building, hybrid fusion, reranking, and citation-grounded LLM synthesis (`gpt-4o-mini`).
2. **Dynamic Knowledge Graph Validity & Traversal Engine:** We implement query-time overlay joining and precedence sorting algorithms that enforce legal hierarchy and temporal validity without requiring rigid database re-indexing.
3. **Multi-Reranker Benchmarking & Optimization:** We provide an extensive empirical comparison across four state-of-the-art cross-encoders, introducing optimized FP16 batching and monkey-patch solutions for seamless Kaggle GPU execution.

1.4 Report Structure

The remainder of this report is organized as follows: Section 2 reviews related work in RAG, Legal IR, hybrid search, and Knowledge-Graph-augmented systems. Section 3 presents the overall L_RAG system architecture and describes each component (retrieval, knowledge graph, reranking, generation) in detail. Section 4 details the Vietnamese legal corpus, knowledge graph schema, and the 500-query benchmark dataset. Section 5 through Section 8 (in subsequent chapters) define the experimental protocol, present ablation results, and analyze system trade-offs.

2 Related Work

This section reviews existing literature and foundational paradigms relevant to Retrieval-Augmented Generation, domain-specific Legal Information Retrieval, hybrid retrieval strategies, and Knowledge-Graph-augmented systems.

2.1 Retrieval-Augmented Generation (RAG)

Retrieval-Augmented Generation (Lewis et al., 2020) has emerged as a dominant framework to mitigate hallucinations, overcome outdated parameter knowledge, and ground Large Language Models (LLMs) in verifiable external knowledge. Standard RAG architectures operate in two main steps: (1) retrieving relevant document chunks from a vector store using dense bi-encoder embeddings, and (2) concatenating retrieved chunks into the prompt context for an LLM to generate an answer.

Despite their success in general-domain benchmarks, baseline RAG architectures face severe limitations in specialized, highly technical domains. In unstructured domain corpora, dense retrieval frequently suffers from low precision due to semantic ambiguity, while fixed-size text chunking loses structural dependencies spanning across paragraphs or document chapters. Advanced RAG paradigms—such as Self-RAG (Asai et al., 2024) and Corrective RAG (CRAG) (Yan et al., 2024)—introduce self-reflection and web-search fallback mechanisms, but they do not explicitly model formal domain taxonomies or statutory legal validity.

2.2 Legal Information Retrieval

Legal Information Retrieval (Legal IR) is a specialized subfield at the intersection of Information Retrieval and Legal Informatics (Locke and Zuccon, 2022). Legal IR differs fundamentally from open-domain text search due to four intrinsic characteristics of legal text:

1. **Hierarchical Structure:** Legal statutes are organized into strict logical trees (Code $\rightarrow$ Part $\rightarrow$ Chapter $\rightarrow$ Section $\rightarrow$ Article $\rightarrow$ Clause $\rightarrow$ Item).

2. **High Precision Requirements:** A single misplaced word or omission of an exception clause can alter the legal interpretation completely.

3. **Statutory Precedence:** Legal systems follow formal legal hierarchies (e.g., the Constitution overrides National Assembly Laws, which in turn override Government Decrees and Ministry Circulars).

4. **Dynamic Validity & Amendments:** Legal provisions are constantly updated, amended, or repealed by subsequent legal instruments over time.

Early Legal IR research focused on rule-based citation parsing and expert systems. Recent approaches leverage specialized transformer models, such as Legal-BERT (Chalkidis et al., 2020) and Law-Former (Xiao et al., 2021), to encode statutory texts. However, relying solely on neural embeddings remains insufficient for capturing complex statutory relationships and temporal applicability.

2.3 Hybrid Retrieval and Reranking Fusion

To overcome the vocabulary mismatch problem of dense embeddings and the semantic rigidity of sparse keyword search, hybrid retrieval combines both modalities (Cormack et al., 2009). Sparse algorithms (e.g., BM25) excel at exact keyword matching—critical for capturing specific legal codes, numbers, and technical terminology—whereas dense bi-encoders (e.g., E5 (Wang et al., 2022)) excel at conceptual paraphrasing.

Combining retrieved lists from heterogeneous engines requires robust rank fusion. Reciprocal Rank Fusion (RRF) (Cormack et al., 2009) merges ranked outputs based on positional reciprocal scores:

$$\mathrm{RRF}(d) = \sum_{m \in M} \frac{1}{k + r_m(d)} \tag{1}$$

where M is the set of retrieval systems, $r_m(d)$ is the rank of document d in system m, and k is a smoothing constant (typically $k = 60$).

Following initial retrieval, cross-encoder rerankers (Nogueira and Cho, 2019) evaluate full joint attention between the query and candidate passages (q, d). Unlike bi-encoders, which compute separate vectors, cross-encoders achieve significantly higher ranking precision (measured via MRR and nDCG) at the expense of higher inference latency. Recent state-of-the-art multilingual rerankers include BGE-Reranker-v2-m3 (Chen et al., 2024), Jina-Reranker-v2 (Jina AI, 2024), and Qwen3-Reranker (Qwen Team, 2025).

2.4 Knowledge-Graph-Augmented Retrieval (GraphRAG)

Knowledge Graphs (KGs) represent domain entities and their explicit relationships as structured nodes and edges (e_1, r, e_2). GraphRAG (Edge et al., 2024) integrates graph traversal with vector search, enabling LLMs to reason over multi-hop relationships and hierarchical structures.

In the legal domain, GraphRAG approaches construct statutory graphs linking documents, provisions, definitions, and legal concepts. At query time, graph traversal algorithms (e.g., Breadth-First Search) expand seed vector hits by navigating structural edges (`HAS_PROVISION`) and citation edges (`CITES`, `AMENDS`). This guarantees that retrieved contexts maintain statutory completeness and legal coherence.

3 System Overview: L_RAG

This section details the end-to-end architecture of **L_RAG**, a Knowledge-Graph-augmented hybrid RAG system engineered for legal information retrieval and question answering over Vietnamese legal corpora.

3.1 Architecture

The L_RAG pipeline unifies structured graph processing, hybrid vector-sparse search, cross-encoder candidate reranking, and citation-grounded LLM synthesis. Figure 1 illustrates the system dataflow across four distinct operational phases: (1) Corpus Ingestion & Graph/Vector Indexing, (2) Multi-Modal Hybrid Retrieval, (3) Knowledge Graph Overlay & Context Expansion, (4) Sequential Candidate Reranking, and (5) LLM Prompting & Generation.

3.2 Retrieval Components

The retrieval module combines three complementary retrieval approaches:

3.2.1 Dense Vector Search

Dense retrieval is powered by `intfloat/multilingual-e5-large`, a 560M parameter transformer embedding model. To solve the structural blindness of standalone chunks, each text chunk is preprocessed into a **Metadata-Enriched Representation**:

$$\begin{aligned}\text{Text}_{\text{embedded}} = \text{Doc\_Title} \parallel \text{Provision\_Header} \\ \parallel \text{Chunk\_Content}\end{aligned} \tag{2}$$

Chunks are stored in an optimized FAISS vector database using `IndexFlatIP` with L2-normalized embeddings for fast cosine similarity search.

3.2.2 Sparse Keyword Search (BM25)

Sparse retrieval utilizes an inverted index built with `rank_bm25`. Raw legal texts are tokenized using the `underthesea` Vietnamese morphological tokenizer, ensuring compound legal terms (e.g., *"bất_động_sản"*, *"trách_nhiệm_hình_sự"*) are indexed as unified lexical tokens. BM25 parameters are set to standard defaults ($k_1 = 1.5, b = 0.75$).

3.2.3 Knowledge Graph Overlay (G-LRAG Module)

The Knowledge Graph module acts as a query coordinator that generates dynamic document filter whitelists (`id_str_filter`) and performs horizontal reading-order expansions. Key features include:

- **Dynamic Overlay Folding:** Validity timelines are joined at query time relative to an `as_of_date` parameter, ensuring superseded provisions are excluded.
- **Precedence Sorting:** Conflicts between legal authority ranks are resolved using statutory hierarchy rules:

$$\begin{aligned}\text{Hiến pháp} \succ \text{Bộ luật / Luật} \\ \succ \text{Nghị định} \succ \text{Thông tư}\end{aligned} \tag{3}$$

- **Reading-Order Traversal:** Structural edge pointers (`CHUNK_NEXT`, `PROVISION_NEXT`) slide context windows horizontally across sibling chunks within the same parent provision.

3.2.4 Fusion Layer

Search hit lists from dense search (R_{dense}) and sparse search (R_{BM25}) are merged using Reciprocal Rank Fusion (RRF) with constant $k = 60$. In configurations without RRF, score normalization is applied using min-max scaling prior to rank merging.

3.3 Reranking Components

Initial retrieval selects $K = 30$ candidate chunks per query. To maximize precision, candidate lists pass through a specialized Cross-Encoder Reranker that scores the full $(query, chunk\_text)$ pair.

L_RAG supports four distinct reranking backends loaded sequentially onto GPU:

1. `mMiniLMv2`: `cross-encoder/mmarco-mMiniLMv2-L12-H384-v1` (Lightweight baseline, 117M params).
2. `BGE-Reranker-v2-m3`: `BAAI/bge-reranker-v2-m3` (Multilingual 567M params with sigmoid activation).
3. `Jina-Reranker-v2`: `jinaai/jina-reranker-v2-base-multilingual` (278M params, featuring dynamic position embedding patches).
4. `Qwen3-Reranker`: `Qwen/Qwen3-Reranker-0.6B` (CausalLM-based 0.6B parameter model evaluated in FP16 precision).

To prevent GPU Out-Of-Memory (OOM) failures during multi-model evaluation, the loader executes a strict **Sequential Load-Run-Unload Cycle**:

$$\begin{aligned}\text{Load}(M_i) \longrightarrow \text{Execute}(M_i) \\ \longrightarrow \text{Delete}(M_i) \longrightarrow \texttt{empty\_cache}()\end{aligned} \tag{4}$$

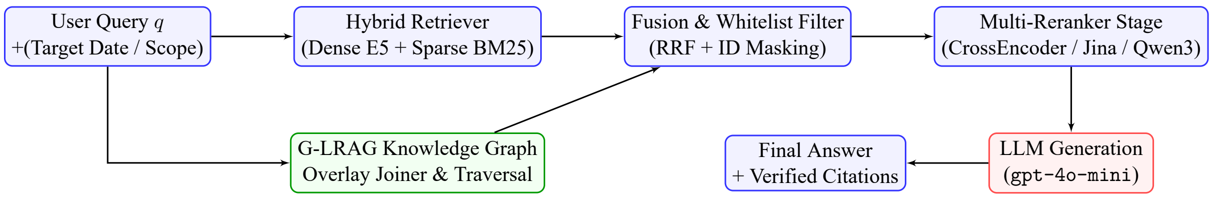

Figure 1: End-to-end operational workflow of the L_RAG system.

3.4 Generation Component

The top $N = 10$ reranked chunks are formatted into a structured prompt context supplied to `gpt-4o-mini` (temperature $T = 0.0$, max tokens 1024).

The system prompt enforces three critical compliance rules:

1. **Strict Citation Grounding:** Answers must explicitly cite governing statutory sources (e.g., *"Theo Điều 15, Luật Đất đai 2024..."*).
2. **Unanswerable Abstention:** If the retrieved context lacks sufficient evidence, the LLM must output `INSUFFICIENT_CONTEXT` rather than hallucination.
3. **Answer-Type Formatting:** Outputs are formatted according to the query type (*boolean*, *extractive*, *abstractive*, or *unanswerable*).

4 Dataset and Benchmark

This section describes the Vietnamese legal corpus, the structural design of the G-LRAG Knowledge Graph, and the 500-query legal QA benchmark dataset used to evaluate the L_RAG system.

4.1 Corpus Description

The legal dataset comprises a massive collection of Vietnamese legal instruments, including National Assembly Laws, Government Decrees, Prime Minister Decisions, and Ministry Circulars. Raw legal documents undergo structured parsing to preserve document metadata, hierarchical provisions, and passage chunks.

Table 1 summarizes the core scale and statistics of the processed corpus.

Each passage chunk is enriched with metadata prior to indexing:

$$\begin{aligned} \text{Chunk\_Meta} = \{&\text{chunk\_id}, \\ &\text{doc\_id}, \text{provision\_id}, \\ &\text{title}, \text{issuing\_authority}, \\ &\text{effective\_date}\} \end{aligned} \tag{5}$$

4.2 Knowledge Graph Structure (G-LRAG Schema)

The G-LRAG Knowledge Graph models statutory entities as nodes and structural and legal dependencies as typed directed edges.

4.2.1 Node Schema

- **Document Node (D):** Represents a full legal instrument. Attributes: `doc_id`, `title`, `authority_rank`, `issuing_body`, `effective_date`, `validity_status`.
- **Provision Node (P):** Represents a logical statutory section (Article/Clause). Attributes: `provision_id`, `header`, `provision_type`.
- **Chunk Node (C):** Represents an indexed passage. Attributes: `chunk_id`, `text`, `sequence_num`.

4.2.2 Edge Schema

Relationships are divided into structural hierarchy and inter-document legal relationships:

- **Structural Hierarchy Edges:**
  - `DOCUMENT_HAS_PROVISION`: Connects Document D to child Provision P.
  - `PROVISION_HAS_CHUNK`: Connects Provision P to child Chunk C.
  - `CHUNK_NEXT`: Sequentially links adjacent sibling chunks $C_i \to C_{i+1}$ within reading order.
  - `PROVISION_NEXT`: Sequentially links adjacent provisions $P_j \to P_{j+1}$.
- **Legal Relationship Edges:**
  - `AMENDS`: Indicates document D_A amends provision/document D_B.
  - `REPLACES`: Indicates document D_A replaces document D_B.
  - `CITES`: Indicates document D_A references document D_B.

Table 1: Overall Statistics of the Vietnamese Legal Corpus.

Entity Type	Description	Total Count
Documents	Full legal instruments (Laws, Decrees, etc.)	151,624
Provisions	Hierarchical units (Articles, Clauses, Items)	1,386,267
Chunks	Fixed/Structural text passages for retrieval	1,513,376
Document Edges	Relationships (Amend, Replace, Cite, Repeal)	883,256
External Stubs	Referenced external entities	19,763

4.2.3 Quarantine & Verification Integrity Gating

To prevent graph poisoning, G-LRAG enforces two strict integrity rules:

1. **Quarantine Gating:** Unverified or draft records are quarantined during ingestion.

2. **Verified-Only Traversal:** Path traversals only traverse edges where `direction_verified == True`.

4.3 QA Benchmark Dataset

To evaluate legal retrieval and answer generation, we constructed a benchmark of **500 Vietnamese legal questions** from the processed G-LRAG corpus using a grounded synthetic QA pipeline.

Questions were generated with `gemini-3.1-flash-lite` from context bundles containing approximately 3–5 related provisions and chunks. The model was instructed to formulate realistic Vietnamese legal questions using only the supplied evidence. Each candidate was subsequently evaluated in a separate verification pass for answer correctness, evidence sufficiency, hallucination avoidance, category correctness, and difficulty correctness. Only candidates passing all verification criteria were retained.

The final generation run sampled 515 documents and 1,697 provisions and achieved a verifier pass rate of 72.41%. After diversity filtering and deduplication, the final benchmark contains **500 questions**: 265 single-hop, 106 citation, 96 legal-validity, 21 multi-hop, and 12 cross-document questions.

The dataset contains **400 answerable** and **100 unanswerable** questions. Answer types include abstractive, boolean, extractive, and unanswerable responses. Unanswerable questions are intentionally constructed without supporting corpus evidence to evaluate the system's ability to abstain from unsupported answers.

4.4 Ground Truth Construction

Each benchmark query contains the question, category, answer type, reference answer, and evidence provenance:

$$Q_i = \{\text{qa\_id}, \text{question}, \text{category}, \\ \text{answer\_type}, \text{reference\_answer}, \text{ground\_truth}\}. \quad (6)$$

The ground-truth object contains the supporting `document_ids`, `provision_ids`, and `chunk_ids`. For answerable questions, the gold chunk identifiers are used to evaluate retrieval with Recall@k, Hit@k, MRR@k, and nDCG@k. For unanswerable questions, the ground-truth evidence sets are empty by design.

5 Experimental Setup

We evaluate L_RAG with controlled, paired ablations over the same Vietnamese legal QA benchmark. The benchmark contains 500 questions: 400 answerable and 100 unanswerable. Runs within an ablation family use the same benchmark, question identifiers, generator settings, and corpus/index version; only the component named by the family is changed. This design is important because the four completed families use different candidate budgets, as summarized in Table 2.

5.1 Research Questions

We organize the experiments around five questions:

1. Does adding document and provision metadata to the embedding text improve exact evidence retrieval or document-level coverage?

2. Does hybrid dense–sparse retrieval improve coverage or ranking over dense retrieval alone?

3. Does the current G-LRAG graph overlay improve evidence coverage and downstream behavior for the tested legal QA benchmark?

4. Which reranker best trades ranking quality against inference cost when the candidate pool is held fixed?

5. Which generator reasoning strategy should be used? The completed generator-reasoning comparison contrasts base prompting with explicit chain-of-thought (CoT) prompting under a fixed retrieved context.

5.2 Metrics

For answerable questions, retrieval is evaluated against the annotated gold chunk IDs using Recall@k, Hit@k, MRR@k, nDCG@k, and Precision@k. For graph runs we additionally report document-identity precision, because a graph variant can return fewer chunks and therefore make a fixed-denominator precision score misleading. The reranker analysis reports candidate Recall@30 and candidate Hit@30 to distinguish the evidence ceiling from the ability of a reranker to promote evidence already present in the candidate pool.

For generated answers, we report exact match, token-level F1, and ROUGE-L on answerable questions. We also report the template-based answerability decision matrix (including non-refusal on answerable questions) and call its aggregate "decision accuracy" rather than treating it as a legal-safety metric. These automatic overlap and abstention measures are diagnostics rather than legal judgments: the available artifacts contain no human labels for legal correctness, completeness, claim faithfulness, or citation-to-chunk alignment. Those metrics are consequently not inferred from lexical overlap.

Efficiency is reported as mean and median per-query latency, with p95 values where the source run recorded them. The reranker runs used cached BM25 candidates and therefore report zero sparse-stage time; those values are not treated as full-online hybrid latency.

5.3 Evaluation Protocol

All completed runs use `intfloat/multilingual-e5-large` for dense retrieval, temperature 0.0, and a maximum of 1,024 output tokens. The non-generator ablations use `gpt-4o-mini`; the reasoning family varies the generator model and prompt mode as listed in Table 2. The embedding-text ablation uses raw FAISS top-10 retrieval with no sparse, graph, fusion, or reranking module. The dense/hybrid/graph ablation retrieves 50 seeds and supplies at most 10 final chunks; graph variants expand five seeds by one hop with a maximum graph context of 10. The reranker ablation reconstructs a shared RRF candidate pool of 30 chunks (`C30`) and compares the top 10 final chunks (`T10`). Its fusion constant is `rrf_k=60`.

Comparisons are paired by `qa_id`. Overall retrieval deltas use the 400 answerable questions. We report percentile bootstrap 95% confidence intervals for the mean per-question delta, using 10,000 resamples for the main comparisons and 2,000 resamples for the smaller reranker slices. A confidence interval containing zero is described as inconclusive rather than as evidence of an effect. The reranker candidate reconstruction agrees with the recorded RRF-only top-10 prefix for all 500 questions.

The generator-reasoning family holds the retrieval stack and intended retrieved top-10 context fixed for each question, and changes the prompt mode for the generator. The primary paired comparison uses the complete GPT-4o-mini Base and CoT runs (500/500 questions, with 400 answerable pairs); their saved contexts match exactly. The DeepSeek-V3.1-Thinking, GLM-5, and Qwen3-8B pairs are also complete but remain descriptive cross-model checks. Four of the 500 shared DeepSeek cases have different exact top-10 chunk-ID sequences between Base and CoT. The Qwen Base and CoT manifests use different code snapshots, timeouts, and retry limits, so their latency and failure comparisons are descriptive. These conditions are not used for the primary paired claim. The post-hoc analysis and frozen result ledger are available at `ablation_study/e2e_LLM_reasoning/results/report.md` and `ablation_work/e2e_LLM_Reasoning/analyze_posthoc.py`.

5.4 Evaluation Workflow and Derived Metrics

Each condition is a saved 500-case run over the same question IDs, benchmark labels, retrieved top-10 context, generation settings, and reference answers. The post-hoc analysis reads these records and does not call a model or rerun retrieval. A row is valid when it is not marked failed, errored, or skipped, has no saved error, and contains a prediction. Lexical metrics use successful answerable rows; a paired Base/CoT statistic uses only IDs that satisfy these conditions in both runs and contain all three lexical metrics.

The original evaluator normalizes text with Unicode NFC, lowercasing, punctuation-to-space replacement, and whitespace collapse. Exact Match is the binary equality of normalized prediction and reference (for boolean items, the first non-empty line is compared). Token F1 uses multiset token overlap, with $F_1 = 2PR/(P + R)$, where P and R are token precision and recall. ROUGE-L uses the token-level longest common subsequence in the same F-score form. For each paired ID we calculate $\Delta = m_{\text{CoT}} - m_{\text{Base}}$, average the deltas, count CoT wins, ties, and losses, and form a percentile paired-bootstrap 95% confidence interval from 10,000 resamples (seed 42). Exact Match also uses an exact McNemar test on discordant binary outcomes.

Additional diagnostics are calculated as follows. Evidence availability is context-recall@$k = |G \cap R_k|/|G|$, where G is the gold chunk-ID set and R_k is the retrieved top-k set; answerable cases are stratified as fully retrieved, partial, or absent. The persisted `unanswerable_accuracy` detector is mapped to answer/refuse decisions: an answerable refusal is an error, while a refusal marker on an unanswerable case is correct. Scheduled decision accuracy uses all 500 rows, whereas valid-only accuracy excludes failed rows. Citation presence is the fraction of successful answerable rows with at least one marker; structural coverage is the mean saved share of scored factual sentences with a marker; invalid-marker rate is total invalid markers divided by all markers. Completion is successful rows divided by scheduled rows, and latency p50/p95 are the saved millisecond values converted to seconds. These fields are structural or operational diagnostics, not tests of citation entailment, legal correctness, or hidden reasoning.

5.5 Ablation Configurations

Table 2 lists the frozen configurations used in the completed analyses. Names are aligned with the run manifests and the derived reports under `ablation_work/`.

5.6 Reproducibility and Artifact Ledger

Each run is accompanied by a manifest, per-question outputs, metric summary, and latency file. The derived analyses are versioned alongside the source outputs: the embedding comparison is in `ablation_study/chunkOnly_vs_chunkMetadata/results/`, the dense/hybrid/graph comparison is in `ablation_study/dense_vs_Hybrid_vs_Graph/results/`, and the multi-reranker analysis is in `ablation_work/reranker_choices/evaluation_runs/ablation3_reranker_v2/`. The script that reconstructs the reranker candidate pool and produces its paired analysis is `ablation_work/reranker_choices/analyze_reranker_metrics.py`.

6 Ablation Study

We report four completed ablation families. The first isolates the text fed to the dense encoder, the second compares dense, hybrid, and graph-augmented retrieval, the third compares six ranking controls over a shared hybrid candidate pool, and the fourth compares generator prompt modes with the retrieved context held fixed. Because candidate construction differs between families, all claims are made within a family; the results are not treated as one pooled factorial experiment.

6.1 Embedding Text: Chunk-Only versus Chunk+Metadata

This comparison uses the same dense encoder, generator, benchmark, seed, and raw FAISS evaluation path. Only the indexed text changes from the chunk body to a header-plus-chunk representation. On 400 paired answerable questions, metadata leaves exact chunk Recall@10 effectively unchanged (48.5% versus 48.5%) but increases document-level success from 65.8% to 70.2%. The paired document-success gain is +4.5 percentage points (95% CI [+0.5,+8.8]), whereas the MRR@10 change is -5.9 points (95% CI $[-9.1, -2.7]$). Metadata therefore improves identification of the correct document while making the first exact gold chunk appear later. This document–provision divergence motivates reporting both views of retrieval; the full comparison is in Table 4.

6.2 Retrieval Mode and Graph Expansion

The second family evaluates DenseOnly, HybridOnly, DenseGraph, and HybridGraph. Hybrid retrieval adds BM25 and RRF to the same dense index; graph variants expand five seed chunks by one hop and retain at most 10 chunks. Without graph expansion, DenseOnly outperforms HybridOnly on Recall@10 (48.55% vs. 42.57%) and nDCG@10 (0.2813 vs. 0.2638). The paired

Table 2: Completed ablation configurations. A dash means that the component was disabled. Candidate budgets are shown because they differ by ablation family.

Family	Configuration	Dense text	Sparse	Graph	Reranking	Candidate/final
Embedding	Chunk-only	chunk text	–	–	–	10/10
Embedding	Chunk+metadata	header + chunk	–	–	–	10/10
Retrieval/graph	DenseOnly	header + chunk	–	–	–	50/10
Retrieval/graph	HybridOnly	header + chunk	BM25	–	–	50/10
Retrieval/graph	DenseGraph	header + chunk	–	G-LRAG	–	50/10
Retrieval/graph	HybridGraph	header + chunk	BM25	G-LRAG	–	50/10
Reranker	None	shared RRF pool	BM25	–	none	30/10
Reranker	RRF-only	shared RRF pool	BM25	–	RRF	30/10
Reranker	mMiniLM	shared RRF pool	BM25	–	mMiniLM CE	30/10
Reranker	Qwen3	shared RRF pool	BM25	–	Qwen3	30/10
Reranker	BGE	shared RRF pool	BM25	–	BGE-v2-m3	30/10
Reranker	Jina	shared RRF pool	BM25	–	Jina-v2	30/10
LLM reasoning	GPT-4o-mini Base	fixed context	–	–	Base prompt	fixed/10
LLM reasoning	GPT-4o-mini CoT	fixed context	–	–	CoT prompt	fixed/10

Recall@10 difference is -5.98 points (95% CI $[-9.71, -2.21]$).

Graph expansion currently displaces more evidence than it adds. Relative to DenseOnly, DenseGraph changes Recall@10 by -12.40 points and loses at least one previously retrieved gold chunk for 61 questions while adding a graph-only gold chunk for only one question. The corresponding HybridOnly to HybridGraph change is -11.81 points, with 59 questions losing a gold chunk and six gaining one. Graph contexts are denser because they contain fewer chunks on average (5.605 for DenseGraph and 6.305 for HybridGraph), but their context recall is lower. The tested graph overlay is thus not yet evidence of a retrieval-quality improvement.

6.3 Reranker Choice at a Fixed Candidate Pool

The third family compares no reranking, RRF-only, mMiniLM, Qwen3, BGE, and Jina over a reconstructed shared RRF pool of 30 candidates and a final top 10. All six runs cover the same 500 questions without retrieval or generation errors, and the reconstruction matches the recorded RRF-only top-10 prefix for 500/500 questions. Candidate Recall@30 is 0.5995 and Candidate Hit@30 is 0.6450 for every configuration, so the comparison measures ranking and retention within a fixed evidence ceiling.

Jina is the strongest retrieval configuration: relative to RRF-only, it improves Recall@10 by +9.23 points, MRR@10 by +13.35 points, and nDCG@10 by +11.85 points; all three 95% paired-bootstrap intervals exclude zero. Qwen3 and BGE also have positive intervals, while mMiniLM supplies a smaller but statistically positive gain at the lowest reranker cost. BGE has the highest token F1 and ROUGE-L, whereas Jina has the highest observed unanswerable accuracy. These answer metrics are lexical/abstention diagnostics, not legal correctness judgments. Table 6 reports the aggregate comparison and Table 7 reports paired effects.

6.4 LLM Generator Ablation: Base versus CoT

This ablation asks whether a reasoning-oriented prompt changes observable answer behavior when the model receives the same retrieved legal material. Every condition uses the same 500-question benchmark (400 answerable and 100 unanswerable), top-10 context, dense–graph–RRF–mMiniLM retrieval stack, temperature 0, and 1,024-token output limit. GPT-4o-mini Base versus CoT is the pre-specified primary comparison. Both runs completed all 500 cases, their 400 answerable rows pair exactly by question ID, and their saved ordered top-10 chunk-ID sequences have zero mismatches.

The post-hoc analysis reads the frozen predictions, aggregate metrics, latency, error, and manifest files; it does not rerun retrieval or generation. Lexical metrics use successful answerable pairs. Deltas are CoT minus Base, with 95% percentile confidence intervals from 10,000 paired bootstrap resamples (seed 42); exact match also receives an exact McNemar test. Answerability decisions use all 500 scheduled cases. These procedures and denominators follow the operative no-rerun metric plan and are detailed in Section 5.4.

6.4.1 Primary paired result

For the primary pair, CoT lowers Token F1 from 24.72% to 23.15% (-1.57 percentage points; 95% CI $[-2.24, -0.92]$) and ROUGE-L from 21.60% to 19.90% (-1.71 points; 95% CI $[-2.34, -1.10]$); both intervals exclude zero. Exact match changes by only -0.25 points and its interval includes zero (McNemar $p = 1.00$). CoT wins/ties/loses on Token F1 in 79/164/157 cases, so the aggregate decline is broad but not universal. It also adds 0.76 seconds to median generation and end-to-end latency and 1.21 seconds to end-to-end p95 latency.

6.4.2 Behavior conditional on available evidence

Because retrieval is fixed, Recall/MRR/nDCG are controls rather than LLM outcomes. We instead stratify answerable cases by saved context Recall@10. Token F1 is lower under CoT when all gold support is retrieved (-1.98 points, $n = 138$), when support is partial (-1.86, $n = 29$), and when no gold support is retrieved (-1.30, $n = 233$). ROUGE-L changes in the same direction (-2.30, -1.59, and -1.37 points). Exact match rises by 0.72 points only in the fully retrieved group, but this small descriptive change does not reverse the primary result. Evidence availability is a context property, not a measure of whether generated claims are faithful to that evidence.

6.4.3 Answerability decisions

The saved field formerly called `unanswerable_accuracy` is not refusal accuracy. It is a template-based answer/refuse decision check over both answerable and unanswerable questions. CoT does not improve recognition of the 100 unanswerable cases: both prompts refuse 98 and answer 2. Its 1.40-point overall increase comes entirely from answering seven more answerable cases, which reduces template-detected false refusals from 85 to 78. The artifacts do not preserve a separate clarification decision, so this is not a semantic measure of refusal helpfulness or appropriateness.

6.4.4 Citation structure, reliability, and descriptive model checks

Table 3 provides one consolidated comparison for all four models and both prompt modes. For the primary GPT pair, CoT raises citation presence from 78.25% to 81.00% and mean unique sources from 1.46 to 1.68, while structural citation coverage falls from 89.42% to 87.91%. Both runs have zero invalid markers. These fields test citation formatting only: they do not establish that a passage entails the adjacent claim.

The other model pairs are descriptive checks rather than additional primary tests. Their CoT-minus-Base Token-F1 changes are -1.33 points for DeepSeek-V3.1-Thinking (95% CI $[-1.96, -0.73]$), -0.68 for GLM-5 ($[-1.23, -0.11]$), and -1.92 for Qwen3-8B ($[-2.77, -1.15]$). Four DeepSeek cases have different ordered top-10 chunk sequences across prompts. The Qwen manifests use different code snapshots, timeout values, and retry limits, so especially its latency and failure comparison is not strictly controlled. No prompt effect is pooled across models.

Claim boundary. The historical artifacts contain final answers, lexical scores, retrieved context, structural citation fields, and latency, but no legal-domain correctness annotations, atomic-claim support labels, citation-entailment judgments, structured public justifications, raw CoT, token accounting, or preferred answer/clarify/refuse labels. Consequently, this ablation supports a bounded conclusion about observable output behavior: GPT-4o-mini CoT reduces template-detected false refusals and increases citation presence, but lowers lexical overlap in every evidence-availability stratum and adds latency. It does not demonstrate better or worse latent reasoning, semantic grounding, legal correctness, citation entailment, or refusal quality. The derived report and machine-readable artifacts are linked from the reproducibility ledger in Section 5.6.

Table 3: Consolidated LLM generator ablation. EM, Token F1, ROUGE-L, and citation values use the 400 answerable outputs; template-based decision accuracy uses all 500 cases. Latencies are p50/p95 seconds. Structural coverage measures citation-marker placement, not semantic entailment.

Model	Prompt	EM	F1	R-L	Decision	Cite pres.	Struct. cov.	Sources	Gen. p50/p95	Total p50/p95
GPT-4o-mini	Base	14.00	24.72	21.60	82.60%	78.25%	89.42%	1.46	2.92/5.74	13.99/17.21
GPT-4o-mini	CoT	13.75	23.15	19.90	84.00%	81.00%	87.91%	1.68	3.68/6.17	14.74/18.42
DeepSeek-V3.1-Thinking	Base	22.25	17.84	15.28	92.00%	90.00%	82.38%	2.57	7.28/15.51	16.15/24.55
DeepSeek-V3.1-Thinking	CoT	22.25	16.51	14.12	92.20%	90.75%	75.21%	2.71	7.82/21.02	18.52/31.37
GLM-5	Base	25.25	16.91	14.90	91.80%	91.75%	80.26%	2.72	4.17/8.79	14.29/19.09
GLM-5	CoT	25.00	16.23	14.27	93.40%	93.75%	78.12%	3.05	4.15/8.73	15.96/21.23
Qwen3-8B	Base	19.25	21.98	18.92	95.80%	97.75%	65.78%	1.79	4.77/9.13	14.15/18.51
Qwen3-8B	CoT	17.25	20.05	17.17	95.20%	98.25%	58.15%	1.81	4.97/10.47	14.58/20.16

7 Results

The completed retrieval, reranking, category, and efficiency results are summarized in the tables below. The generator-reasoning ablation, including its paired results and evidence-stratified analysis, is reported in Section 6.4. Retrieval metrics use answerable questions and are paired within each experiment family. We use percentage points (pp) for deltas so that the magnitude of an effect is not confused with a relative percentage change.

7.1 Embedding Representation

Table 4 shows that adding metadata to the embedded text changes the level at which retrieval succeeds. At top 10, exact chunk Recall is unchanged to one decimal place, while document success increases by 4.5 pp. The paired MRR decrease is statistically distinguishable from zero, so metadata should not be described as an unqualified improvement: it favors document identification but does not improve the rank of the first exact gold chunk.

7.2 Dense, Hybrid, and Graph Retrieval

The DenseOnly configuration is the strongest exact-chunk retriever in the second family. HybridOnly is lower on Recall@10 and nDCG@10, and its paired Recall loss has a 95% interval entirely below zero. Graph expansion lowers Recall@10 for both seed retrievers: by 12.40 pp for DenseOnly and 11.81 pp for HybridOnly. The graph-only evidence counts show why: graph expansion adds a gold chunk for one DenseOnly query and six HybridOnly queries, but removes at least one previously retrieved gold chunk for 61 and 59 queries, respectively.

The denser graph contexts improve ID precision (0.0580 to 0.0800 for Dense and 0.0493 to 0.0641 for Hybrid), but this should be read together with the large context-recall loss. Graph expansion also increases recorded unanswerable accuracy from 0.38 to 0.41 for Dense and from 0.38 to 0.47 for Hybrid, while paired Token F1 changes are small and their confidence intervals include zero. The graph implementation therefore produces a precision–coverage trade-off rather than a demonstrated overall gain.

7.3 Reranker Comparison

Table 6 reports the six configurations evaluated on the fixed `C30` candidate pool. Jina obtains the strongest retrieval scores, while BGE has the highest overlap scores. Qwen3 is competitive in quality but incurs a much larger reranker-stage cost. The recorded unanswerable-accuracy column is retained from the source metrics; semantic legal-answer evaluation is not available.

All rerankers have positive paired retrieval intervals for the three primary ranking measures. In contrast, every false-refusal interval includes zero. Jina also retains 0.8850 of the gold chunks available in `C30`, versus 0.7129 for RRF-only, confirming that its gain comes from better promotion of available evidence rather than a larger candidate pool. The full safety and win/tie/loss analysis is stored with the derived artifacts.

7.4 Category and Answer-Type Analysis

The largest Jina gains occur for legal-validity and citation questions, while the smaller extractive slice is inconclusive. Table 8 reports the requested Jina-versus-RRF comparison; all slices are answerable cases, and intervals use 2,000 paired bootstrap resamples. The multi-hop slice has only 20 questions and is exploratory. The hard difficulty and cross-document slices are also too small to support a main conclusion.

7.5 Generation and Efficiency

The deterministic answer metrics do not establish legal correctness. In the embedding family, metadata changes Token F1 by +0.1 pp and ROUGE-

Table 4: Embedding-text ablation on 400 paired answerable questions. Values are percentages; "document success" means that a retrieved chunk from the gold document is present in the indicated context. Deltas are chunk+metadata minus chunk-only.

Variant	Hit@5	Recall@5	MRR@5	Doc.@5	Hit@10	Recall@10	MRR@10	Doc.@10
Chunk-only	44.0	39.0	27.4	59.0	53.5	48.5	28.6	65.8
Chunk+metadata	40.2	36.8	20.9	61.8	53.2	48.5	22.7	70.2
Delta	-3.8	-2.2	-6.5	$+2.8$	-0.2	$+0.1$	-5.9	$+4.5$

Table 5: Dense/hybrid retrieval and graph results. Retrieval values use 400 answerable questions. "ID precision" divides the number of gold chunks by the actual number of returned chunks; p50/p95 latency is in seconds over all 500 questions.

Method	Recall@10	MRR@10	nDCG@10	ID precision	Avg. chunks	Retrieval p50/p95	Total p50/p95
DenseOnly	0.4855	0.2267	0.2813	0.0580	10.000	0.797/0.856	5.483/9.946
HybridOnly	0.4257	0.2252	0.2638	0.0493	10.000	27.855/27.924	31.538/33.623
DenseGraph	0.3615	0.2045	0.2374	0.0800	5.605	0.798/0.860	4.998/9.750
HybridGraph	0.3076	0.2012	0.2228	0.0641	6.305	27.858/27.929	31.180/33.312

L by -0.1 pp. In the dense/hybrid/graph family, graph expansion changes Token F1 by approximately +0.4 pp for Dense and +0.2 pp for Hybrid, with bootstrap intervals crossing zero. In the reranker family, BGE has the strongest observed overlap scores, but Jina has the best retrieval scores and a lower total p50 than BGE (4.158 s vs. 5.848 s). The results support a conditional engineering choice, not a claim that one reranker produces more legally correct answers.

Efficiency differences are substantial. Hybrid retrieval raises total median latency from 5.483 s to 31.538 s and reports a one-time BM25 precomputation cost of about 13,495 s. Among rerankers, mMiniLM is fastest (0.385 s reranker-stage p50), Jina is intermediate (1.278 s), BGE is slower (2.872 s), and Qwen3 is much slower (9.695 s). The reranker artifacts do not record p95, peak GPU memory, or full-online sparse latency, so those deployment quantities remain open.

8 Discussion

8.1 Interpretation of Findings

The ablations separate four effects that are often conflated in a legal RAG pipeline. First, metadata-enriched embeddings improve document-level identification without improving exact provision ranking. The representation therefore helps the system locate the right instrument, but the additional header tokens can compete with the passage text when the evaluation target is a specific gold chunk.

Second, the current hybrid implementation does not improve exact-chunk retrieval over the dense baseline. BM25 and RRF may still supply context that is marginally useful to generation, but the paired retrieval loss and large latency increase make hybrid retrieval a poor default under the present settings. This result should not be generalized to all fusion methods: it identifies a limitation of this index, tokenization, fusion constant, and candidate budget.

Third, graph expansion improves identity precision by returning fewer, denser contexts, but it removes more already-retrieved gold evidence than it adds. The likely mechanism is a fixed context budget combined with one-hop expansion, which replaces seed passages with structurally adjacent passages. Graph context can still be valuable for validity and citation reasoning, but that hypothesis requires semantic legal-answer annotations and a graph policy that preserves high-confidence seed evidence.

Finally, reranking is the clearest positive result. With the candidate pool held fixed, every tested cross-encoder improves the three primary ranking metrics over RRF-only, and Jina has the largest paired gains. This is evidence for better ordering and candidate retention, not evidence that the generator's legal claims are correct. The quality–latency frontier favors Jina for the current benchmark, BGE when lexical answer overlap is prioritized, and mMiniLM when latency is the primary constraint.

The generator-reasoning comparison gives a different kind of result. With the retrieved context fixed, GPT-4o-mini CoT lowers Token F1 and

Table 6: Multi-reranker ablation over a shared 30-candidate pool and final top-10 context. Retrieval and overlap values are means over 400 answerable questions; recorded unanswerable accuracy and p50 latency use the complete 500-case runs.

Model	Recall@10	Hit@10	MRR@10	nDCG@10	Token F1	ROUGE-L	Rec. unans. acc.	Rerank p50	Total p50
None	0.4842	0.5375	0.2167	0.2714	0.2801	0.2453	0.856	0.000	2.925
RRF-only	0.4382	0.4800	0.2335	0.2728	0.2750	0.2416	0.874	0.000	2.972
mMiniLM	0.4973	0.5425	0.3047	0.3425	0.2901	0.2568	0.872	0.385	3.343
Qwen3	0.5218	0.5725	0.3418	0.3688	0.2912	0.2583	0.876	9.695	13.007
BGE	0.5188	0.5675	0.3247	0.3593	**0.2955**	**0.2603**	0.868	2.872	5.848
Jina	**0.5305**	**0.5850**	**0.3670**	**0.3914**	0.2913	0.2558	**0.884**	1.278	**4.158**

Table 7: Paired effects of each reranker relative to RRF-only. Deltas are in percentage points; intervals are 95% percentile bootstrap CIs from 10,000 question-level resamples over the 400 answerable cases. A negative false-refusal delta indicates fewer refusals on answerable questions.

Reranker	Δ Recall@10	Δ MRR@10	Δ nDCG@10	Δ false refusal
mMiniLM	$+5.91$ $[+1.85, +9.96]$	$+7.12$ $[+3.55, +10.68]$	$+6.96$ $[+3.77, +10.26]$	$+0.25$ $[-3.00, +3.50]$
Qwen3	$+8.37$ $[+4.57, +12.25]$	$+10.84$ $[+7.24, +14.36]$	$+9.60$ $[+6.52, +12.73]$	-0.25 $[-3.25, +2.75]$
BGE	$+8.07$ $[+4.29, +11.80]$	$+9.12$ $[+5.50, +12.70]$	$+8.65$ $[+5.47, +11.84]$	$+0.75$ $[-2.25, +3.75]$
Jina	$+9.23$ $[+5.51, +12.98]$	$+13.35$ $[+9.51, +17.24]$	$+11.85$ $[+8.46, +15.27]$	-1.25 $[-4.00, +1.25]$

ROUGE-L relative to Base prompting, with paired intervals excluding zero, while exact match is nearly unchanged. CoT reduces template-detected false refusals and increases citation-marker presence, but it also increases generation and total latency and slightly lowers structural citation coverage. Thus explicit reasoning is not a demonstrated quality improvement for this prompt, model, and benchmark; its observed benefit is limited to output-format diagnostics that lack semantic legal validation. The DeepSeek, GLM, and Qwen pairs are complete but descriptive. The Qwen Base/CoT manifests use different code snapshots, timeout values, and retry limits, and four shared DeepSeek cases have different exact top-10 chunk-ID sequences, so cross-model and fixed-context claims outside the GPT primary pair should remain cautious.

8.2 Design Implications

For a retrieval-focused deployment, we recommend dense retrieval with a reranker, using Jina as the current default candidate and mMiniLM as the latency-oriented fallback. BGE is a reasonable alternative when its observed Token F1 and ROUGE-L advantage is preferred. Hybrid retrieval should be revisited after measuring online BM25 cost and tuning fusion; the current results do not justify paying its reported latency by default. Graph augmentation should be gated by a validity- or relationship-aware query mode until seed-preserving expansion and semantic evaluation are completed.

These recommendations are conditional. A production legal system should select evidence using validity metadata, expose citations, abstain when evidence is insufficient, and require human verification. No automatic metric reported here can certify the legal effect of an answer.

8.3 Threats to Validity

The benchmark has 500 questions and a small number of multi-hop, hard, and cross-document cases. Ground-truth chunk sets may not capture every passage that a legal expert would consider sufficient, and document-level success is not equivalent to provision-level correctness. All runs use one embedding model and one corpus/index version within each family; generator settings are fixed within each compared cell, while the reasoning family intentionally varies the prompt mode. The source artifacts also differ in candidate budget across families, which prevents a pooled comparison of absolute scores.

The reranker and generator-reasoning safety analyses use deterministic refusal-marker/template classifiers, not human assessment. The reasoning artifacts also contain no raw CoT, reasoning-token counts, cost, or semantic claim annotations. The DeepSeek pair is complete, but four shared cases have exact retrieved-context sequence differences. The GLM and Qwen pairs are complete but descriptive; Qwen Base and CoT also differ in code snapshot, timeout, and retry configuration. The cached BM25 stage means recorded reranker latency is not full-online hybrid latency, and p95 latency, peak GPU memory, and

Table 8: Jina versus RRF-only on selected category and answer-type slices. Deltas are percentage points with 95% paired-bootstrap CIs.

Dimension	Slice	n	△ Recall@10	△ MRR@10	△ nDCG@10
Category	citation	106	$+11.89$ [$+2.82$, $+21.70$]	$+16.22$ [$+7.86$, $+24.51$]	$+15.82$ [$+7.51$, $+23.82$]
Category	legal validity	91	$+14.47$ [$+6.96$, $+21.43$]	$+18.90$ [$+10.28$, $+27.26$]	$+15.22$ [$+9.13$, $+21.50$]
Category	single hop	172	$+5.04$ [-0.00, $+10.37$]	$+9.63$ [$+4.34$, $+14.69$]	$+8.41$ [$+3.81$, $+13.05$]
Category	multi hop	20	$+7.50$ [-0.83, $+17.50$]	$+10.12$ [-3.42, $+24.33$]	$+7.95$ [-0.77, $+17.98$]
Answer type	extractive	78	$+0.13$ [-8.85, $+9.74$]	$+8.53$ [-0.90, $+17.39$]	$+7.14$ [-1.33, $+16.23$]
Answer type	abstractive	199	$+13.65$ [$+8.50$, $+18.80$]	$+14.74$ [$+9.61$, $+20.14$]	$+13.81$ [$+9.56$, $+18.71$]
Answer type	boolean	123	$+7.86$ [$+1.63$, $+14.50$]	$+14.18$ [$+7.64$, $+21.13$]	$+11.68$ [$+6.27$, $+17.95$]

controlled throughput are unavailable. Finally, the graph corpus contains unresolved relationship-direction and validity-annotation questions; these may affect the behavior of graph-augmented runs. We therefore present the results as benchmark-bounded engineering evidence rather than as a claim of legal reliability.

Limitations

The corpus may contain missing body text, incomplete metadata, external references, and relationship-direction ambiguity. The in-memory graph also has a substantial memory footprint at the scale of the current corpus. Retrieval and generation quality are bounded by the benchmark's coverage and ground-truth annotations, and a fluent answer is not evidence of legal correctness. The system is therefore a research prototype and must not be used as a substitute for qualified legal review. The reasoning analysis is also limited to saved output records: it has no semantic claim labels, raw chain-of-thought, reasoning-token accounting, or cost data. The DeepSeek pair is complete, but four shared cases have different exact top-10 chunk-ID sequences between Base and CoT. The Qwen3-8B Base/CoT pair is complete, but its manifests use different code snapshots, timeout values, and retry limits; latency and failure comparisons for that pair are therefore descriptive.

Ethical Considerations

Legal QA can affect access to rights, benefits, and remedies. The system should expose source citations, validity metadata, and abstention behavior; users should verify the cited instrument and its current status before taking action. Evaluation and deployment should also account for uneven corpus coverage, language variation, privacy in user queries, and the risk of over-trusting automated legal explanations.

9 Conclusion

We presented a controlled ablation study of L_RAG for Vietnamese legal QA. Metadata-enriched embeddings improved document-level retrieval but not exact chunk ranking. The tested hybrid fusion reduced exact-chunk retrieval and added substantial latency, while one-hop graph expansion increased context density at the cost of coverage. The multi-reranker study produced the most consistent gain: at a fixed candidate pool, Jina improved Recall@10, MRR@10, and nDCG@10 over RRF-only with paired confidence intervals above zero. BGE led lexical answer overlap and mMiniLM offered the lowest reranking cost. The completed generator-reasoning ablation shows that GPT-4o-mini CoT prompting lowers Token F1 by 1.57 pp and ROUGE-L by 1.71 pp relative to Base prompting under fixed retrieved context, while improving only template-based answerability diagnostics and adding latency.

These findings support dense retrieval followed by a cross-encoder as the current engineering direction, with Jina as the quality–latency candidate for this benchmark. They do not establish legal correctness, citation faithfulness, or safe deployment because the available runs lack validated semantic legal-answer annotations.

The refreshed reasoning inventory also shows why run-manifest identity and completion checks are necessary before cross-model prompt conclusions.

9.1 Future Work

The next experiments should collect blinded labels for legal correctness, completeness, claim faithfulness, citation precision, and citation recall; measure full-online p95 latency, GPU memory, and throughput; and validate relationship direction against authoritative document pairs. Additional work includes a candidate-pool sweep, RRF and graph-budget tuning, seed-preserving graph expansion, investigation of the four DeepSeek context-order mismatches, provenance-aligned reruns for the Qwen3-8B pair, and evaluation on a larger and more diverse Vietnamese legal benchmark.

Acknowledgments

The authors thank Dr. Le Thanh Tung for guidance on the course project and the contributors who prepared the Vietnamese legal benchmark and evaluation artifacts.

References

Akari Asai, Zeqiu Wu, Yizhong Wang, Avirup Sil, and Hannaneh Hajishirzi. 2024. Self-RAG: Learning to retrieve, generate, and critique through self-reflection. In *International Conference on Learning Representations*.

Ilias Chalkidis, Manos Fergadiotis, Prodromos Malakasiotis, Nikolaos Aletras, and Ion Androutsopoulos. 2020. LEGAL-BERT: The muppets straight out of law school. In *Findings of the Association for Computational Linguistics: EMNLP 2020*, pages 2898–2904.

Jianlv Chen, Shufang Xiao, et al. 2024. BGE M3-embedding: Multi-lingual, multi-functionality, multi-granularity text embeddings through self-knowledge distillation. arXiv preprint arXiv:2402.03216.

Gordon V. Cormack, Charles L. A. Clarke, and Stefan Buettcher. 2009. Reciprocal rank fusion outperforms condorcet and individual rank learning methods. In *Proceedings of the 32nd International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 758–759.

Douglas Edge, Ha Trinh, Newey Cheng, Joshua Bradley, Alex Chao, Apurva Mody, Robert Truitt, and Jonathan Larson. 2024. From local to global: A graph RAG approach to query-focused summarization. *arXiv preprint arXiv:2404.16130*.

Jina AI. 2024. Jina reranker v2 for agentic RAG. Technical report.

Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Kuttler, Mike Lewis, Wen-tau Yih, Tim Rocktaschel, Sebastian Riedel, and Douwe Kiela. 2020. Retrieval-augmented generation for knowledge-intensive nlp tasks. In *Advances in Neural Information Processing Systems*.

Daniel Locke and Guido Zuccon. 2022. Case law retrieval: Problems, methods, challenges and evaluations in the last 20 years. *arXiv preprint arXiv:2202.07209*.

Rodrigo Nogueira and Kyungha Cho. 2019. Passage re-ranking with BERT. *arXiv preprint arXiv:1901.04085*.

Qwen Team. 2025. Qwen3 embedding: Advancing text embedding and reranking through foundation models. arXiv preprint arXiv:2506.05176.

Liang Wang, Nan Yang, Xiaolong Huang, Binbin Jiao, et al. 2022. Text and code embeddings by contrastive pre-training. *arXiv preprint arXiv:2201.10005*.

Chaojun Xiao, Xueyu Zhong, Zijun Zhang, et al. 2021. Lawformer: A pre-trained language model for chinese legal long documents. *arXiv preprint arXiv:2105.03887*.

Shiyu Yan et al. 2024. Corrective retrieval augmented generation. *arXiv preprint arXiv:2401.15884*.

A Dataset and Graph Schema

The corpus represents legal instruments as documents, hierarchical provisions, and indexed chunks. A document stores its identifier, title, issuing authority, effective-date and validity metadata; a provision stores its article/clause hierarchy; and a chunk stores passage text, parent identifiers, and reading order. The graph uses containment edges `DOCUMENT_HAS_PROVISION` and `PROVISION_HAS_CHUNK`, reading-order edges `CHUNK_NEXT` and `PROVISION_NEXT`, and document-level legal relationships such as amendment, replacement, repeal, and citation. The completed evaluation uses the chunk identifiers in the QA gold sets as the retrieval relevance units.

The processed corpus summary contains 151,624 documents, 1,386,267 provisions, 1,513,376 chunks, 883,256 document relationships, and 19,763 external stubs. The relationship and validity fields require direction validation before they are used to make high-stakes legal filtering decisions.

B Additional Results

For the reranker family, candidate Recall@30 and candidate Hit@30 are 0.5995 and 0.6450 for every configuration. Gold retention@10 is 0.7129 for RRF-only, 0.8256 for mMiniLM, 0.8702 for Qwen3, 0.8630 for BGE, and 0.8850 for Jina. The source-derived Markdown report additionally records paired win/tie/loss counts, all requested slices, lexical safety diagnostics, and the status of the blinded annotation template. These values are retained in the machine-readable JSON rather than duplicated as a long appendix table.

Table 9: Effect of graph expansion on top-10 gold evidence. "Graph-only" is gold evidence introduced by the graph context; "graph-missed" is gold evidence present before expansion but absent afterward. Counts are over 400 answerable questions.

Base	Δ Recall@10	Δ MRR@10	Δ nDCG@10	Graph-only	Graph-missed
DenseOnly → DenseGraph	−0.1240	−0.0222	−0.0439	1 query (0.25%)	61 queries (15.25%)
HybridOnly → HybridGraph	−0.1181	−0.0240	−0.0411	6 queries (1.50%)	59 queries (14.75%)

C Run Manifests and Artifacts

All paths below are relative to the repository root. The lean commit retains manifests, aggregate metrics, reports, and latency summaries. Raw per-question outputs, retrieval caches, and the pending annotation template remain local evaluation artifacts and are intentionally not versioned here.

- Embedding text: `ablation_work/chunkOnly_vs_chunkMetadata/`
- Dense/hybrid/graph: `ablation_work/denseOnly_vs_denseGraph/` and `ablation_work/hybridOnly_vs_hybridGraph/`
- Reranker source and derived analysis: `ablation_work/reranker_choices/evaluation_runs/ablation3_reranker_v2/`
- Reranker analysis script: `ablation_work/reranker_choices/analyze_reranker_metrics.py`
- Generator-reasoning source runs and derived report: `ablation_work/e2e_LLM_Reasoning/` and `ablation_study/e2e_LLM_reasoning/results/`
- Generator-reasoning analysis script: `ablation_work/e2e_LLM_Reasoning/analyze_posthoc.py`
- Human-evaluation scaffold: generated locally for the pending semantic-evaluation phase and omitted from the lean commit.

The formal report's Markdown counterparts are organized under `ablation_study/`. The derived reports intentionally distinguish completed deterministic analyses from unavailable semantic legal-judge metrics and unmeasured full-online deployment statistics.